CSE-322: Mid-Semester Examination

Winter-2026

February 22, 2026

Full Marks: 60 **Time: 2:00 hours**

General Instructions: This is a closed book examination. Please be precise, brief and to the point in your answer. Unnecessary blabbers or unreadable handwriting will fetch deduction of marks.

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1. Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA and b be particular symbol of Σ, such that for all $q \in Q$, $\delta(q, b) = q$. Prove or disprove that: $\{b\}^* \subseteq L(M)$ or $\{b\}^* \cap L(M) = \emptyset$. **(8 Marks)**

   Solution: The statement is TRUE.

   Proof: We crucially need the fact that $\widehat{\delta}(q_0, b^n) = q_0$ for any $n \geq 0$. We prove by induction on n that for every $n \geq 0$, $\widehat{\delta}(q_0, b^n) = q_0$.

   Base case: When $n = 0$. Then, $\widehat{\delta}(q_0, \varepsilon) = q_0$. Hence the base case is true.

   Induction hypothesis: Assume that the statement is true for every $n \leq k$.

   Induction Step: Consider $\widehat{\delta}(q_0, b^{k+1})$.

   Note that $\widehat{\delta}(q_0, b^{k+1}) = \delta(\widehat{\delta}(q_0, b^k), b)$.

   Due to induction hypothesis, $\widehat{\delta}(q_0, b^k) = q_0$ and by definition, $\delta(q_0, b) = q_0$.

   Hence, $\widehat{\delta}(q_0, b^{k+1}) = q_0$.

   There are 2 cases.

   - Case-(i): Let $q_0 \in F$.
     Then, for every $n \geq 0$, $\widehat{\delta}(q_0, b^n) = q_0 \in F$.
     Then, for every $n \geq 0$, $b^n \in L(M)$.
     Then, $\{b\}^* \subseteq L(M)$.
   - Case-(ii): Let $q_0 \notin F$.
     Since for every $n \geq 0$, $\widehat{\delta}(q_0, b^n) = q_0$ and $q_0 \notin F$.
     Therefore, $\widehat{\delta}(q_0, b^n) \notin F$.
     Therefore, for every $n \geq 0$, $b^n \notin L(M)$.
     In this case, $\{b\}^* \cap L(M) = \emptyset$.

   Since the above cases are mutually exhaustive, therefore $\{b\}^* \subseteq L(M)$ or $\{b\}^* \cap L(M) = \emptyset$.

2. Consider two languages A and B. Define $\mathsf{Rquo}(A, B) = \{x \in \Sigma^* \mid \text{there exists } y \in B \text{ such that } xy \in A\}$. Prove or disprove that $\mathsf{Rquo}(A, B) \cdot B = A$. **(8 Marks)**

   Solution: The statement is FALSE.

   Counterexample: Consider the two languages $A = \{ab, aba\}$ and $B = \{a\}$.

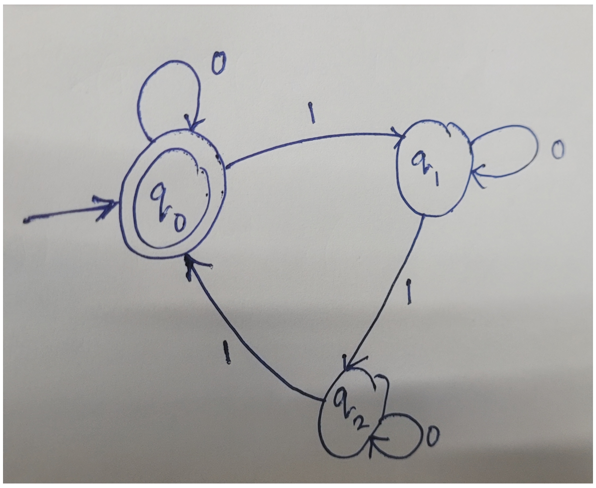
   Then, $\mathsf{Rquo}(A, B) = \{ab\}$.

Then, $\mathsf{Rquo}(A,B)\cdot B = \{aba\}$.

Note that both A and B are regular languages, and $\mathsf{Rquo}(A,B)\cdot B \neq A$.

3. Design a DFA M that accepts $B = \{w \in \{0,1\}^* \mid \text{the number of 1's in } w \text{ is multiple of } 3\}$. Formally prove that $L(M) = B$. **(10 Marks)**

Solution: The transition diagram of the DFA $M = (Q, \Sigma, \delta, q_0, F)$ is as follows.



Clearly, $F = \{q_0\}$.

We prove that $B = L(M)$.

Proof: There are two directions.

($\subseteq$) Let $w \in B$.

Base Case: $|w| = 0$ as $\varepsilon \in B$.

Then, $\widehat{\delta}(q_0, \varepsilon) = q_0 \in F$.

Hence, $w \in L(M)$.

Induction Hypothesis: The statement is true for every string $w \in B$ of length at most k.

Induction Step: Consider $w \in B$ such that $|w| = k+1$.

Since $w \in B$, the number of 1s in w a multiple of 3. If w has no 1 at all, then $w = 0^{k+1}$.

Then it trivially follows that $\widehat{\delta}(q_0, w) = q_0 \in F$. Hence, $w \in L(M)$.

Suppose that the number of 1's in w is $3(p+1)$ for some $p \geq 0$.

Then, consider the substring x of w such that $w = xy$, the first symbol of y is 1, and y contains exactly 3 ones.

Then, x contains exactly $3p$ many 1's.

Consider $\widehat{\delta}(q_0, xy)$.

Note that $\widehat{\delta}(q_0, xy) = \widehat{\delta}(\widehat{\delta}(q_0, x), y)$.

Since x contains $3p$ many 1s, hence $x \in B$ and $|x| \leq k$, due to induction hypothesis, $\widehat{\delta}(q_0, x) = \{q_0\}$.

Then, observe that y is of the form $10^a10^b10^c$.

Since $\widehat{\delta}(q_i, 0^a) = q_i$ for any $i = 1, 2, 3$, therefore, $\widehat{\delta}(q_0, 10^a10^b10^c) = q_0$.

As $q_0 \in F$, therefore, $w \in L(M)$.

Therefore, $B \subseteq L(M)$.

($\supseteq$) Let $w \in L(M)$.

Base Case: $|w| = 0$ since $\varepsilon \in L(M)$. Observe that $w \in B$ by definition of B. Hence, $L(M) \subseteq B$.

Induction Hypothesis: For every $w \in L(M)$ such that $|w| \leq k$, $w \in B$.

Consider $w \in L(M)$ such that $|w| = k + 1$.

Consider $w = xy$ such that y is the smallest nomempty string satisfying that $x \in L(M)$.

If y has no 1, then $\widehat{\delta}(q_0, 0^a) = q_0$. Due to induction hypothesis, $|x| \leq k$ and $\widehat{\delta}(q_0, x) = q_0$.

Then, $x \in B$.

Since y has no 1, therefore the number of 1's in w is a multiple of 3.

Suppose that y has at least one 1.

Then, y has exactly 3 ones.

Similarly, due to induction hypothesis, $|x| \leq k$ and $x \in B$.

Then, $\widehat{\delta}(q_0, xy) = \widehat{\delta}(q_0, y)$.

As y has exactly three 1's, y is of the form 10^a10^b.

Then, $\widehat{\delta}(q_0, y) = q_0 = \widehat{\delta}(q_0, w)$.

Therefore, the number of 1's in $xy = w$ is a multiple of 3.

Hence, $w \in B$.

4. Let A and B be two regular languages. Then, we define $\text{Tsh}(A, B) = \{a_1b_1c_1 \ldots a_mb_mc_m \mid a_1 \ldots a_m \in A, b_1c_1 \ldots b_mc_m \in B, a_i, b_i, c_i \in \Sigma \text{ for all } 1 \leq i \leq m\}$.

   Prove or disprove that $\mathsf{Tsh}(A, B)$ is regular. **(9 Marks)**

   Solution: The statement is TRUE.

   Proof: As A and B are regular languages, suppose that $M_A = (Q_A, \Sigma, \delta_A, q_A, F_A)$ and $M_B = (Q_B, \Sigma, \delta_B, q_B, F_B)$ are the DFAs satisfying that $L(M_A) = A$ and $L(M_B) = B$.

   - **DFA Construction:** We construct a DFA $M_D = (Q_D, \Sigma, \delta_D, q_D, F_D)$ as follows.
     – $Q_D = Q_A \times Q_B \times \{1, 2, 3\}$.
     – $q_D = (q_A, q_B, 1)$.
     – $\delta_D : Q_D \times \Sigma \to Q_D$ is defined as
       $\delta_D((p_A, p_B, 1), a) = (\delta_A(p_A, a), p_B, 2)$.
       $\delta_D((p_A, p_B, 2), a) = (p_A, \delta_B(p_B, a), 3)$.
       $\delta_D((p_A, p_B, 3), a) = (p_A, \delta_B(p_B, a), 1)$.
     – $F_D = F_A \times F_B \times \{1\}$.
   - **Correctness:** The correctness argument involves proving that $L(M_D) = \mathsf{Tsh}(A, B)$.
     Assume that $w = a_1b_1c_1 \ldots a_mb_mc_m$. Then, it is crucial to prove that

     $$\widehat{\delta_D}((p_A, p_B, 1), w) = (\widehat{\delta_A}(p_A, a_1 \ldots a_m), \widehat{\delta_B}(p_B, b_1c_1 \ldots b_mc_m), 1)$$

     We prove this by induction on m.
     Base Case: $m = 0$ and $m = 1$. Then, $a_1 \ldots a_m = \varepsilon$ and $b_1c_1 \ldots b_mc_m = \varepsilon$.
     Then, the statement is trivial.
     $\widehat{\delta_D}((p_A, p_B, 1), a_1b_1c_1)$ is computed as follows.
     $\delta_D((p_A, p_B, 1), a_1) = (\delta_A(p_A, a_1), p_B, 2)$.
     Then, $\delta_D((\delta_A(p_A, a_1), p_B, 1), b_1, 2) = (\delta_A(p_A, a_1), \delta_B(p_B, b), 3)$ and
     $\delta_D((\delta_A(p_A, a_1), \delta_B(p_B, b_1), 3), c_1) = (\delta_A(p_A, a_1), \widehat{\delta_B}(p_B, b_1c_1), 1)$.
     Hence, $\widehat{\delta_D}((p_A, p_B, 1), a_1b_1c_1) = (\widehat{\delta_A}(p_A, a_1), \widehat{\delta_B}(p_B, b_1c_1), 1)$.
     Suppose that the statement is true for every $m \leq k$.

Consider $m = k + 1$.
Then, $w = xa_{k+1}b_{k+1}c_{k+1}$ such that $|x| = 3k$ and $x = a_1b_1c_1 \dots a_kb_kc_k$.
For the sake of simplicity, let us use $x_1 = a_1 \dots a_k$ and $y_1 = b_1c_1 \dots b_kc_k$.
Then, due to induction hypothesis $\widehat{\delta_D}((p_A, p_B, 1), x)) = (\widehat{\delta_A}(p_A, x_1), \widehat{\delta_B}(p_B, y_1), 1)$.
Then, $\widehat{\delta_D}((p_A, p_B, 1), xa_{k+1}b_{k+1}c_{k+1}) = \widehat{\delta_D}(\widehat{\delta_D}((p_A, p_B, 1), x), a_{k+1}b_{k+1}c_{k+1})$.
As $\widehat{\delta_D}((p_A, p_B, 1), x)) = (\widehat{\delta_A}(p_A, x_1), \widehat{\delta_B}(p_B, y_1), 1)$, therefore
$\widehat{\delta_D}(\widehat{\delta_D}((p_A, p_B, 1), x), a_{k+1}b_{k+1}c_{k+1}) = \widehat{\delta_D}((\widehat{\delta_A}(p_A, x_1), \widehat{\delta_B}(p_B, y_1), 1), a_{k+1}b_{k+1}c_{k+1})$.
Then, the RHS of the previous equation equals to $(\widehat{\delta_A}(p_A, x_1a_{k+1}), \widehat{\delta_B}(p_B, y_1b_{k+1}c_{k+1}), 1)$.
Therefore, our claim is true.

Now, we use this to prove that $L(M_D) = \mathsf{Tsh}(A, B)$.

($\supseteq$) Let $w \in \mathsf{Tsh}(A, B)$.
Then, $w = a_1b_1c_1 \dots a_mb_mc_m$ such that $x = a_1a_2 \dots a_m \in A$ and $y = b_1c_1b_2c_2 \dots b_mc_m \in B$.
Then, due to the above claim, $\widehat{\delta_D}((q_A, q_B, 1), w) = (\widehat{\delta_A}(q_A, x), \widehat{\delta_B}(q_B, y), 1)$.
Note that $\widehat{\delta_A}(q_A, x) \in F_A$ and $\widehat{\delta_B}(q_B, y) \in F_B$.
Then, $(\widehat{\delta_A}(q_A, x), \widehat{\delta_B}(q_B, y), 1) \in F_D$.
Hence, $w \in L(M_D)$.

($\subseteq$) Let $w \in L(M_D)$.
Then, $\widehat{\delta_D}((q_A, q_B, 1), w) \in F_A \times F_B \times 1$.
Then, $|w|$ is a multiple of 3.
Suppose that $w = a_1b_1c_1 \dots a_kb_kc_k$.
Due to the earlier claim, $\widehat{\delta_D}((q_A, q_B, 1), w) = (\widehat{\delta_A}(q_A, a_1 \dots a_k), \widehat{\delta_B}(q_B, b_1c_1 \dots b_kc_k), 1)$.
Since $\widehat{\delta_D}((q_A, q_B, 1), w) \in F_A \times F_B \times \{1\}$, it must be that $\widehat{\delta_A}(q_A, a_1 \dots a_k) \in F_A$, and $\widehat{\delta_B}(q_B, b_1c_1 \dots b_kc_k) \in F_B$.
Then, $a_1 \dots a_k \in L(M_A) = A$ and $b_1c_1 \dots b_kc_k \in L(M_B) = B$.
Hence, $a_1b_1c_1 \dots a_kb_kc_k \in \mathsf{Tsh}(A, B)$.

5. Let Σ_1 and Σ_2 denote two disjoint finite alphabets and a function $f : \Sigma_1 \to \Sigma_2^2$. Define $h : \Sigma_1^* \to \Sigma_2^*$ as follows:

   - $h(\varepsilon) = \varepsilon$.
   - $h(w) = f(a_1) \dots f(a_m)$ when $w = a_1 \dots a_m$ such that for all $1 \leq i \leq m$, $a_i \in \Sigma_1$.

   For a language $L \subseteq \Sigma_1^*$, define $\mathrm{Hom}(L) = \{h(w) \mid w \in L\}$.

   If $B \subseteq \Sigma_1^*$ is a regular language such that $M_B = (Q_B, \Sigma_1, \delta_B, q_B, F_B)$ is a DFA satisfying $L(M_B) = B$. Use the description of M_B to construct an explicit NFA $N = (Q_N, \Sigma_2, \delta_N, q_N, F_N)$ such that $L(N) = \mathrm{Hom}(B)$. Formally prove that your NFA is correct.

   (9 Marks)

   Solution: <u>NFA Construction:</u> For the sake of simplicity, assume $q_0 = q_B$ and $Q_B = \{q_0, q_1, \dots, q_n\}$.

   We construct an NFA N as follows.

   - Define $Q_N = Q_B \cup \{p_a^i \mid 0 \leq i \leq n \text{ and } a \in \Sigma_1\}$.
   - For every $a \in \Sigma_1$, we consider $f(a) = x_1x_2$ such that $x_1, x_2 \in \Sigma_2$.
     If $\delta_B(q_i, a) = q_j$, then $\delta_N(q_i, x_1) = p_a^i$ and $\delta_N(p_a^i, x_2) = q_j$.

- $q_N = q_B$.
- $F_N = F_B$.

This completes our construction. Now, we prove that $L(N) = \mathsf{Hom}(B)$.

Observation 1- From the construction of our transition function,

$\forall q_i \in Q_B, x \in \Sigma_2, \delta_N(q_i, x) \cap Q_B = \phi$

Similarly,

$\forall p_a^i \in (Q_N \setminus Q_B), x \in \Sigma_2, \delta_N(p_a^i, x) \cap (Q_N \setminus Q_B) = \phi$

These observations imply that the transition function of N takes the disjoint and exhaustive subsets of the states, namely Q_B and $Q_N \setminus Q_B$ and when applied on a state from one of them, gives a state in the other subset.

Claim 1 -

$$q_j \in \hat{\delta}_N(q_i, f(a)) \iff \delta_B(q_i, a) = q_j$$

proof -

[$\Longrightarrow$] Given $q_j \in \hat{\delta}_N(q_i, f(a))$. Let $f(a) = x_1x_2$ such that $x_1, x_2 \in \Sigma_2$. Therefore, $q_j \in \hat{\delta}_N(q_i, x_1x_2) = \delta_N(\delta_N(q_i, x_1), x_2)$. Let $\delta_N(q_i, x_1) = p$. Therefore since $\delta_N(p, x_2) = q_j$, we can use the construction rule for the transition function and say that $\delta_B(q_i, a) = q_j$.

[$\Longleftarrow$] Given $\delta_B(q_i, a) = q_j$. Let $f(a) = x_1x_2$. Via construction rule of δ_N, $\delta_N(q_i, x_1) = p_a^i$ and $\delta_N(p_a^i, x_2) = q_j$ where $f(a) = x_1x_2$. Therefore, $\hat{\delta}_N(q_i, f(a)) = \hat{\delta}_N(q_i, x_1x_2) = \delta_N(\delta_N(q_i, x_1), x_2) = \delta_N(p_a^i, x_2) = q_j$. Hence we have proved the claim.

($\subseteq$) Let $w = x_1x_2...x_{n-1}x_n \in L(N)$.

Therefore, w has an accepting path such that $\hat{\delta}_N(q_N, w) = q_F$ for some $q_F \in F_N$.

Since $F_N = F_B \subseteq Q_B$, $q_F \in Q_B$. Similarly, $q_N = q_0 \in Q_B$. Therefore our accepting path begins in Q_B and ends in Q_B and alternates between states of Q_B and $Q_N \setminus Q_B$.

Examining the accepting path, we can cut them into pieces of 2 characters. Hence there must exist $m \geq 0$ and symbols $a_1, a_2, ..., a_m \in \Sigma_1$ such that $w = f(a_1)f(a_2)...f(a_m)$. The existence of the pre-image of every pair of characters under f is guaranteed by claim 1 as each piece of 2 characters in our decomposition takes a state to another therefore showing the inverse under f exists.

Because the accepting run in N ends in some $q_f \in F_B$, it must have the form

$$q_B \xrightarrow{f(a_1)} q_{i_1} \xrightarrow{f(a_2)} q_{i_2} \cdots \xrightarrow{f(a_m)} q_f.$$

By the property of the transition function given in the construction, this implies that in M_B we have

$$q_B \xrightarrow{a_1} q_{i_1} \xrightarrow{a_2} q_{i_2} \cdots \xrightarrow{a_m} q_f.$$

Thus $w = a_1 \cdots a_m \in B$. Since $h(w) = f(a_1) \cdots f(a_m) = y$, we obtain $y \in \mathsf{Hom}(B)$. Therefore, $L(N) \subseteq \mathsf{Hom}(B)$

($\supseteq$) Let $w \in \mathsf{Hom}(B)$. Therefore, $\exists y = a_1a_2...a_m \in B$ such that $w = h(y) = f(a_1)f(a_2)...f(a_m)$.

Since $y \in B = L(M_B)$, there exist states in M_B such that $q_B \xrightarrow{a_1} q_{i_1} \xrightarrow{a_2} q_{i_2} \cdots \xrightarrow{a_m} q_f$, and $q_f \in F_B$

Via property of the transition function of N mentioned at construction, we can then say that if $f(a_i) = x_1^i x_2^i$, then since $\delta_B(q_{i_j}, a_i) = q_{i_k}$ for all a_i, then $\delta_N(q_{i_j}, x_1^i) = p_{x_1^i}^{i_j}$ and $\delta_N(p_{x_1^i}^{i_j}, x_2^i) = q_{i_k}$. Therefore, $\hat{\delta}_N(q_{i_j}, f(a_i)) = q_{i_k}$.

Therefore, on input $w = h(y) = f(a_1)f(a_2)...f(a_m)$, we get (in N)

$$q_N = q_B \xrightarrow{f(a_1)} q_{i_1} \xrightarrow{f(a_2)} q_{i_2} \cdots \xrightarrow{f(a_m)} q_f$$

Therefore, $\hat{\delta}(q_N, h(y)) = q_f$ for this path. Since $q_f \in F_N$, we get that N accepts $w = h(y) \implies w \in L(N)$. Thus, $\mathsf{Hom}(B) \subseteq L(N)$.

Hence we have proved that $\mathsf{Hom}(B) = L(N)$

6. Let B denote the set of all strings from $\{0,1\}^*$ whose length is a perfect square. Prove or disprove that B is regular. **(8 Marks)**

Solution: The statement is FALSE because B is not a regular language.

Proof: Suppose for the sake of contradiction that B is regular. We consider an arbitrary constant n.

Consider the string $w = 0^n 1^{n^2+n+1} \in B$.

We consider every possible decomposition $w = xyz$ into substrings.

If $y \neq \varepsilon$, then it violates one of the conditions of pumping lemma.

If $|xy| > n$, then it violates another condition of pumping lemma.

Now, we consider every possible decomposition such that

- $y \neq \emptyset$, and
- $|xy| \leq n$.

Then, $x = 0^p$, $y = 0^q$, and $z = 0^{n-p-q}1^{n^2+n+1}$ such that $p + q \leq n$ and $q > 0$.

In addition, $1 \leq q \leq n$.

Then, we choose $k = 0$ and consider $xy^kz = xz$.

Note that $xz = 0^{n-q}1^{n^2+n+1}$.

Now, observe that $|xz| = n^2 + 2n + 1 - q$.

Since $1 \leq q \leq n$, it must be that $n^2 + n + 1 \leq |xz| \leq n^2 + 2n$.

Observe that $n^2 < |xz| < (n+1)^2$.

Since $|xz|$ appears between two consecutive perfect square numbers, hence $|xz|$ is not a perfect square.

Then, $xz \notin B$, contradicting the properties of pumping lemma.

Therefore, B is not a regular language.

7. Let $B = \{0^{3n}w1^n \mid w \in \{0,1\}^* \text{ and } n \geq 1\}$. Either prove that B is not a regular language, or provide an explicit decomposition of B into 2 or 3 regular languages. In case you provide an explicit decomposition of B into 2 or 3 regular languages, then you must formally prove that your decomposition is correct. **(8 Marks)**

<u>**N.B.**</u> − A language B admits a decomposition into 2 languages if $B = B_1B_2$ such that both B_1 and B_2 are two languages. If you provide a decomposition $B = B_1B_2$, then you must formally prove that $B = B_1B_2$.

Solution: The language B admits a decomposition B into 3 regular languages.

We define

- $A_1 = \{000\}$,
- $A_2 = \{0,1\}^*$, and
- $A_3 = \{1\}$.

Observe that both A_1, A_2, and A_3 are regular languages.

Now, we prove that $B = A_1 \cdot A_2 \cdot A_3$.

($\supseteq$) Let $w \in A_1 \cdot A_2 \cdot A_3$.

Then, $w = xyz$ such that $x = 000$, $y \in \Sigma^*$, and $z = 1$.

Then, observe that w can be represented as 0^3y1 such that $y \in \{0, 1\}^*$.

Hence, $w \in B$.

($\subseteq$) Let $w \in B$.

Then, w is of the form $0^{3n}x1^n$ for some $n \geq 1$ such that $x \in \Sigma^*$.

We consider the first 3 symbols and the last one symbol from w.

Then, we get that $w = 0^3 \cdot 0^{3n-3} \cdot x \cdot 1^{n-1} \cdot 1$.

Then, $0^{3n-3} \cdot x \cdot 1^{n-1} = y \in \{0, 1\}^*$.

Hence, w is of the form 0^3y1.

Observe that $000 \in A_1$, $y \in \{0, 1\}^* = A_2$ and $1 \in A_3$.

Hence, $w \in A_1 \cdot A_2 \cdot A_3$.